

Precision

Definition. Precision refers to the degree of reproducibility or agreement among repeated measurements of the same quantity under unchanged conditions.

Formally, if $x_1, x_2, \dots, x_n$ are independent measurements of a fixed quantity, precision is inversely related to the dispersion of these values, commonly quantified by the sample standard deviation

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}, \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

A small s indicates high precision; a large s indicates low precision. Precision concerns the *spread* of measurements around their own mean and is distinct from accuracy, which concerns the closeness of that mean to the true value μ .

Worked example. Suppose a student measures the period of a pendulum five times, obtaining $T = 2.01, 2.03, 1.99, 2.02, 2.00$ seconds. The mean is $\bar{T} = 2.010$ s. Computing the sample standard deviation yields $s \approx 0.0158$ s. Since the values cluster tightly around $\bar{T}$, the measurements are precise, regardless of whether $\bar{T}$ equals the pendulum's true period.

Key theorem. *Bias–variance decomposition of mean squared error.* For an estimator $\hat{\theta}$ of a parameter θ ,

$$\mathbb{E}[(\hat{\theta} - \theta)^2] = \underbrace{(\mathbb{E}[\hat{\theta}] - \theta)^2}_{\text{bias}^2 \text{ (accuracy)}} + \underbrace{\text{Var}(\hat{\theta})}_{\text{precision}}.$$

This identity formalizes the separation between accuracy (bias) and precision (variance): total measurement error decomposes exactly into these two independent contributions, and eliminating one does not guarantee eliminating the other.

Lab cell (SymPy).

```
import sympy as sp

data = [2.01, 2.03, 1.99, 2.02, 2.00]
```

```
n = len(data)
mean = sp.Rational(sum(data), n).evalf()

variance = sum((sp.Rational(x) - mean)**2 for x in data) / (n - 1)
std_dev = sp.sqrt(variance)

print(f"Mean: {mean}")
print(f"Sample standard deviation (precision measure): {std_dev.evalf()})
```